

# Michael O. Dirksen

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## Contact

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## EDUCATION

**M.S. in Financial Mathematics** – University of Chicago

Expected 2029

*Advanced Scholars Program*

**B.S. in Mathematics** – University of Chicago

2024 - 2028

*GPA: 3.8/4.0*

**High School Diploma** – Lincoln Northeast High School

2020 - 2024

*GPA: 4.5/4.0, Rank: 1/768*

## ACADEMIC EXPERIENCE

### Teaching Assistant

Sep 2025 – Present

Department of Mathematics, University of Chicago

- Led weekly tutorial and problem-solving sessions for the undergraduate calculus sequence (Calculus I–III), emphasizing both foundational proof techniques and computational methods
- Provided individualized instruction and assessment for a cohort of 15 students through grading, office hours, and guided problem sessions
- Completed departmental pedagogy training and assisted with course administration, grading, and instructional support

### Booth Research Assistant

Dec 2024 – Present

Department of Statistics, Chicago Booth School of Business (*Advisor: Claire Tseng*)

- Contribute to an ongoing working paper on large language models and asset pricing, supporting empirical analysis and model evaluation
- Collect, clean, and analyze large-scale financial and prediction market data using Python, with emphasis on statistical inference and significance testing of predictive LLM outputs
- Assist with empirical investigation of options markets, including data-driven analysis of market structure and informational content
- Implement and evaluate machine learning and LLM-based methods via APIs, integrating computational modeling with financial and statistical insights

### Student Researcher

Sep 2025 – Jan 2026

Center for Spatial Data Science, University of Chicago (*Advisor: Luc Anselin*)

- Assisted in the development and evaluation of an AI-based assistant for GeoDa, supporting spatial data analysis workflows
- Conducted systematic testing and parameter evaluation of multiple AI models using spatial datasets, documenting performance and limitations

- Prepared technical notes and analytical feedback for the instructor, informing model refinement and instructional design
- Contributed to the development of a new AI-focused laboratory component for the undergraduate Spatial Analysis course curriculum

## PROFESSIONAL EXPERIENCE

### Quantitative Analyst Intern

Dec 2025 – Present

J. Goldman & Co. L.P. | New York, NY

- Build Python-based research systems to evaluate signals across a \$5bn market-neutral portfolio
- Integrate financial datasets into research pipelines supporting signal allocation and evaluation
- Leverage GPT-5 with structured JSON to standardize trade ideas, signal labeling, and analysis

### Investment Management Intern

May 2025 – Aug 2025

Wespath Institutional Investments | Chicago, IL

- Analyzed FactSet data, contributing to the reallocation of 84% of a \$28bn portfolio
- Presented attribution at 3/9 investment meetings, discussing notes on external stakeholders
- Piloted 4 projects utilizing AI Copilot and Python-based automation to workplace efficiency

### Corporate Advisory Intern

Jan 2024 – Apr 2025

PricewaterhouseCoopers | Chicago, IL

- Selected as 1 of 6 final presenters to pitch industry research to 5 PwC and Indego Africa executives
- Conducted diligence for 5 donor leads, compiled in a 15-slide deliverable for the partner company
- Received a perfect evaluation on my pitch deck, resulting in a \$1,000 stipend for my deliverable

### Real Estate Valuation Intern

Mar 2025 – May 2025

Walker & Dunlop | Remote

- Engaged in sessions with leaders across Walker & Dunlop, gaining insights into market operations
- Completed a course focused on principles in asset management and financial modeling
- Awarded an academic stipend based on performance and engagement throughout the program

## HONORS & AWARDS

**Jane Street GUTS++ (2nd)** – Jane Street

2025

2nd Place in Jane Street Invitational Mathematics Competition

**Jane Morton & Henry C. Murphy Award Nominee** – University of Chicago

2025

Undergraduate nomination for exceptional contributions to the university community

**Humanitarian Award Nominee** – University of Chicago

2025

Undergraduate Nomination for commitment to the welfare of the Hyde Park community

**FEEA Merit Scholarship Recipient** – FEEA

2025

Federal Merit Scholarship for Children of Government Employees

**Believers & Achievers Scholarship** – Currency Capital

2024

Merit Scholarship for Academic & Extracurricular Excellence

**4x Nebraska Academic All-State** – NSAA

2020 – 2024

Award for highest state wide GPA within a Varsity Sport

## FELLOWSHIPS & GRANTS

|                                                                                    |             |
|------------------------------------------------------------------------------------|-------------|
| <b>Advanced Scholars "4+1" Fellowship</b> – University of Chicago                  | 2025 – 2029 |
| Opportunity to Pursue a Master's Degree in a 5th Year at the University of Chicago |             |
| <b>Financial Markets Program</b> – University of Chicago                           | 2025 – 2028 |
| Selective 4-year program at the designed to enhance quantitative finance acumen    |             |
| <b>Metcalf Research Grant</b> – University of Chicago                              | 2020 – 2024 |
| Awarded to support an exploratory research learning opportunity                    |             |

## RELEVANT COURSEWORK

### Undergraduate Courses

The University of Chicago [2024 - 2028]

|                                                              |             |
|--------------------------------------------------------------|-------------|
| MATH 20310 Accelerated Analysis in $\mathbb{R}^n$ II         | Winter 2025 |
| SOSC 13220 Spatial Analysis II                               | Winter 2025 |
| CSMC 14200 Introduction to Computer Science II               | Winter 2025 |
| STAT 24500 Introduction to Statistical Theory and Methods II | Winter 2025 |
| MATH 20310 Accelerated Analysis in $\mathbb{R}^n$ I          | Autumn 2025 |
| STAT 25100 Introduction to Mathematical Probability          | Autumn 2025 |
| SOSC 13120 Spatial Analysis I                                | Autumn 2025 |
| STAT 24400 Introduction to Statistical Theory and Methods I  | Autumn 2025 |
| MATH 16300 Honors Calculus III                               | Spring 2025 |
| PHYS 13300 Waves, Optics, & Heat                             | Spring 2025 |
| MATH 20250 Abstract Linear Algebra                           | Spring 2025 |
| MATH 16200 Honors Calculus II                                | Winter 2025 |
| PHYS 13200 Electricity & Magnetism                           | Winter 2025 |
| MATH 16100 Honors Calculus I                                 | Autumn 2024 |
| PHYS 13100 Mechanics                                         | Autumn 2024 |

## COMMUNITY INVOLVEMENT

**Undergraduate Student Government: Finance Committee** Dec 2024 – Present  
Emergency Funds Chair | Chicago, IL

- Deliberate in meetings to distribute a \$2.5M fund to over 100 students groups in need annually
- Co-headed alumni outreach for the emergency funds board, raising +\$5,000 from alumni
- Worked with the Dean of Students to create modules to advocate for incoming first-year students

**The Derivatives Group** Sep 2024 – Present  
Board Member & Treasurer | Chicago, IL

- Taught two 3-hour courses to 80+ students, covering introductory mathematical finance concepts
- Headed corporate sponsor outreach, acquiring two new sponsors for trading competitions
- Learned about Measure Theory, Brownian Motion, and Stochastic processes over 12 weeks

**UChicago Behavioral Economics Association**

Sep 2024 – Present

VP of Finance &amp; Project Lead | Chicago, IL

- Manage a \$7,000 club budget to organize and fund events to promote education for 70+ members
- Led the strategy team on 2 community consulting projects, deriving solutions for local businesses
- Established a speaker series between the club and Booth's Behavioral Sciences Division Professors

**Student Leadership Advisory Council**

Sep 2024 – Present

Student Ambassador | Chicago, IL

- Develop campus research, using data-based insights to optimize University student opportunities
- Engage in meetings with university board members, advocating for new leadership opportunities
- Allocate \$3M funding to on-campus organizations through student and internal deliberations

**Chicago Public Schools**

Dec 2024 – May 2025

Academic Assistant Intern | Chicago, IL

- Taught a class of 20 pre-college students, covering, algebra, basic numeracy, and problem-solving
- Led reading and literacy seminars, facilitating small-group discussions to improve comprehension
- Provided small-group tutoring in math homework, test preparation, and skill reinforcement

## ACADEMIC INTERESTS

**Mathematics**

Measure Theory, Stochastic Processes, Brownian Motion, Dynamical Systems, Mathematical Billiards, Optimization Theory, and Combinatorial Algorithms

**Statistics**

Nonparametric & Semiparametric Inference, High-Dimensional Statistics, Empirical Process Theory, and Statistical Learning Theory

**Economics / Finance**

Asset Pricing Theory, Econometrics, Game Theory, Stochastic Control and Optimization, Applications of Machine Learning, and Computational Methods for Large-Scale Data